

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Class Test

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 3 – Class Test
Weightage:	20% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
N-gram Models and Word Counting for Language Modeling	N-gram Construction, Word Frequency Analysis, Probability Estimation	1
Unsmoothed N-grams and Smoothing Techniques in Syntax Analysis	Unsmoothed N-grams, Backoff, Deleted Interpolation	2
Entropy and Language Uncertainty in NLP Systems	Information Theory, Entropy Measurement, Language Prediction	3
English Word Classes and Tagsets for Linguistic Analysis	Word Categories, POS Categories, English Tagsets	4
Part-of-Speech Tagging Techniques and Applications	POS Tagging, Rule-Based Tagging, Stochastic Tagging, Transformation-Based Tagging	5

1. Document Processing System

Input Words

- talked
- talking
- talks

Step 1: Morphological Analysis

Word	Root	Suffix	Type
talked	talk	-ed	Inflectional (Past Tense)
talking	talk	-ing	Inflectional (Present Participle)
talks	talk	-s	Inflectional (3rd Person Singular)

Step 2: Porter Stemmer

Word	Rule Applied	Final Stem
talked	Remove -ed	talk
talking	Remove -ing	talk
talks	Remove -s	talk

Step 3: Normalized Output

talked
talking
talks

↓

talk

Role of Morphological Normalization

- Groups all grammatical variations into one base form.
- Reduces duplicate index entries.
- Improves search accuracy and retrieval efficiency.
- Enables users searching for **"talk"** to retrieve documents containing **talked**, **talking**, and **talks**.

2. Hybrid Morphological Processing Pipeline

Input Words

- relational
- conditional
- happiness
- running

Morphological Breakdown

Word	Root	Affix	Type
relational	relate	-ional	Derivational
conditional	condition	-al	Derivational
happiness	happy	-ness	Derivational
running	run	-ing	Inflectional

Finite-State Parsing

relational

START → relate → +ional → relational

conditional

START → condition → +al → conditional

happiness

START → happy → +ness → happiness

running

START → run → +ing → running

Porter Stemmer

Word	Stemming Steps	Final Stem
relational	relational → relate → relat	relat
conditional	conditional → condition	condition
happiness	happiness → happy	happi
running	running → run	run

Normalized Output

Input	Final Stem
relational	relat
conditional	condition
happiness	happi
running	run

Contribution of Each Technique

- **Derivational Analysis:** Identifies suffixes that change word class.
- **Inflectional Analysis:** Removes grammatical endings.
- **Finite-State Parsing:** Validates legal word formation using state transitions.
- **Porter Stemmer:** Produces a common normalized stem for indexing.

3. Enterprise Search Engine Optimization (SEO) System

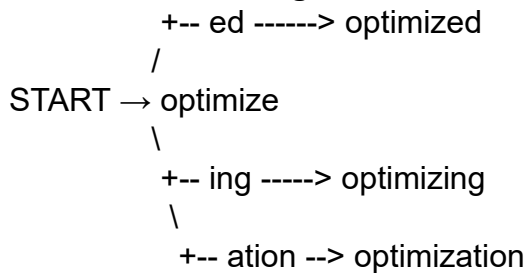
Input Words

- optimize
- optimization
- optimized
- optimizing

Morphological Analysis

Word	Root	Affix	Type
optimize	optimize	—	Base Form
optimization	optimize	-ation	Derivational
optimized	optimize	-ed	Inflectional
optimizing	optimize	-ing	Inflectional

Finite-State Parsing



Porter Stemmer

Word	Final Stem
optimize	optim
optimization	optim
optimized	optim
optimizing	optim

Normalized Output

optimize
optimization
optimized
optimizing
↓
optim

Benefits for SEO

- Groups all keyword variations under one stem.
- Eliminates duplicate indexing.
- Improves keyword matching and ranking.
- Enhances search relevance and reduces storage redundancy.

4. Large-Scale Information Retrieval System

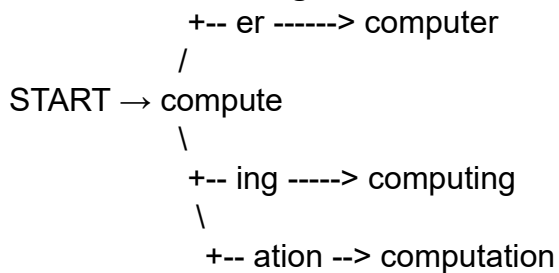
Input Words

- compute
- computer
- computing
- computation

Morphological Breakdown

Word	Root	Affix	Type
compute	compute	—	Base Form
computer	compute	-er	Derivational
computing	compute	-ing	Inflectional
computation	compute	-ation	Derivational

Finite-State Parsing



Porter Stemmer

Word	Final Stem
compute	comput
computer	comput
computing	comput
computation	comput

Final Normalized Output

compute
computer
computing
computation
↓
comput

Final Table

Word	Morphological Structure	State Transition	Final Stem
compute	compute	START → compute	comput
computer	compute + er	compute → er	comput
computing	compute + ing	compute → ing	comput
computation	compute + ation	compute → ation	comput

5. Text Mining System

Input Words

- decide
- decision
- decisive
- deciding

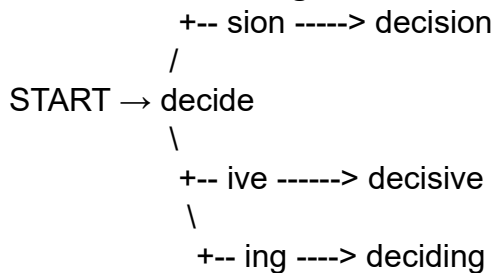
Morphological Breakdown

Word	Root	Affix	Type
decide	decide	—	Base Form
decision	decide	-sion	Derivational
decisive	decide	-ive	Derivational
deciding	decide	-ing	Inflectional

Effect of Suffixes

Suffix	Meaning	Word Class
-sion	Forms a noun	decision
-ive	Forms an adjective	decisive
-ing	Indicates present participle	deciding

Finite-State Parsing



Porter Stemmer

Word	Final Stem
decide	decid
decision	decis
decisive	decis
deciding	decid

Note: Porter stemming may produce slightly different stems (**decid** and **decis**) because it follows rule-based suffix stripping rather than semantic normalization.

Normalized Representation (Morphological Root)

For semantic clustering, all words are mapped to the common root:

decide
decision
decisive
deciding
↓
decide

Final Table

Word	Morphological Structure	State Transition	Porter Stem	Semantic Root
decide	decide	START → decide	decid	decide
decision	decide + sion	decide → sion	decis	decide
decisive	decide + ive	decide → ive	decis	decide
deciding	decide + ing	decide → ing	decid	decide

Faculty In-charge	Course Coordinator	Program Director

